

Combinatorial Solving with Provably Correct Results

Lecture 1

Jakob Nordström

University of Copenhagen and Lund University

Topics in Algorithms and Complexity

University of Copenhagen

October 1, 2026



Brief Presentation

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Research interests:

- Computational complexity theory
- Applied combinatorial optimisation
- Certifying algorithms / proof logging



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Research interests:

- *Proving exponential lower bounds for NP-hard problems*
- *Designing algorithms to solve NP-hard problems in (ideally) linear time*
- *Generating machine-verifiable proofs that such algorithms reason correctly*



Lectures Based on Research Contributions By...

- Berhan Oumer Adame
- Markus Anders
- Jeremias Berg
- Bart Bogaerts
- Benjamin Bogø
- Jordi Coll
- Wolf De Wulf
- Emir Demirović
- Simon Dold
- Jan Elffers
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- Ross McBride
- Ciaran McCreesh
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- Magnus O. Myreen
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Combinatorial Solving and Optimisation

- Revolution last couple of decades in **combinatorial solvers** for
 - Boolean satisfiability (SAT) solving [BHvMW21]¹
 - Constraint programming (CP) [RvBW06]
 - Mixed integer linear programming (MIP) [AW13, BR07]
- Solve NP-complete problems (or worse) very successfully in practice!
- **Except solvers are sometimes wrong...** (Even best commercial ones) [BLB10, CKSW13, AGJ⁺18, GSD19, BMN22, GCS23]
- Solvers can propose infeasible “solutions” (but such errors can be detected)
- More challenging: How to achieve reliable claims of infeasibility?
- Or that a solution is optimal? (Even off-by-one mistakes can snowball into large errors if solver used as subroutine)

¹See end of slides for all references with bibliographic details

What Can Be Done About Solver Bugs?

■ Software testing

Very useful, but bugs slip through even with careful domain-specific testing

Progress using [fuzzing](#) and [delta debugging](#) [BB09, BLB10, KB22, NPB22, PB23]

But testing inherently can only detect presence of bugs, not absence

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■ Formal methods

Prove that solver implementation adheres to formal specification

Current [formal verification](#) techniques cannot scale to level of complexity in modern solvers
(Despite valiant efforts in, e.g., [Fle20])

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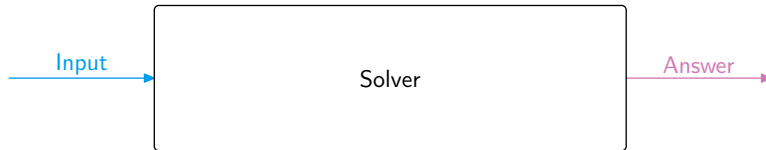
Current [formal verification](#) techniques cannot scale to level of complexity in modern solvers
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■ Proof logging

Make solver [certifying](#) [ABM⁺11, MMNS11] by adding code so that it outputs

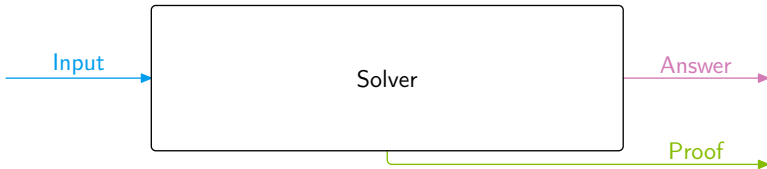
- 1 not only [answer](#) but also
- 2 simple, machine-verifiable [proof](#) that answer is correct

Proof Logging with Certifying Solvers: Workflow



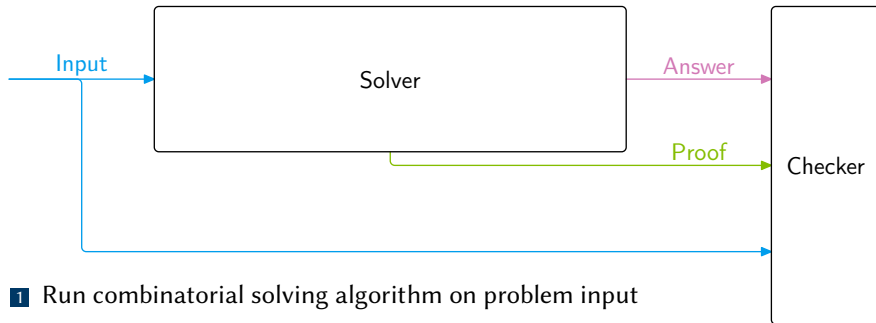
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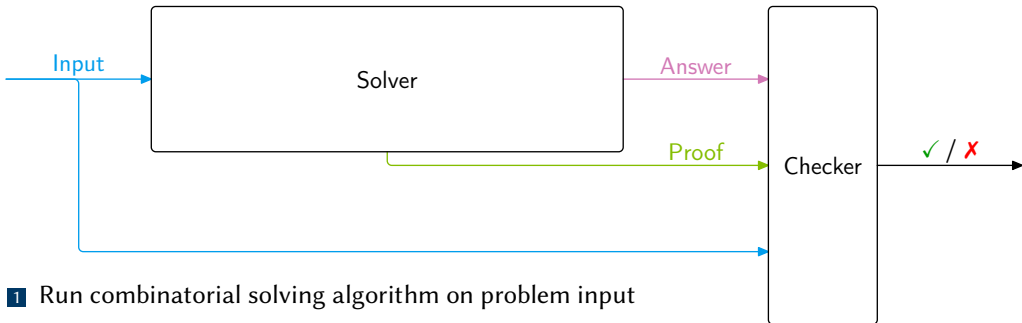
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Proof Logging with Certifying Solvers: Workflow



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- 3 Feed input + answer + proof to proof checker

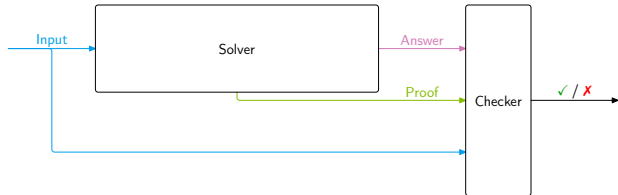
Proof Logging with Certifying Solvers: Workflow



- 1 Run combinatorial solving algorithm on problem input
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- 4 Verify that proof checker says answer is correct

Proof Logging Desiderata

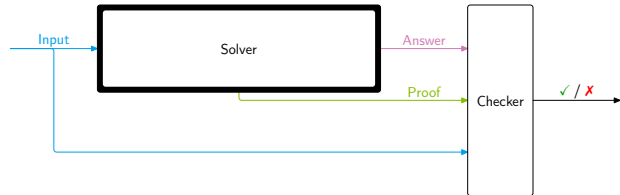
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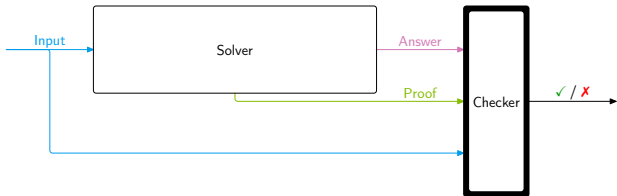
- **very powerful:** minimal overhead for sophisticated reasoning



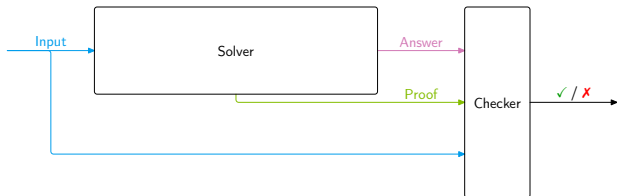
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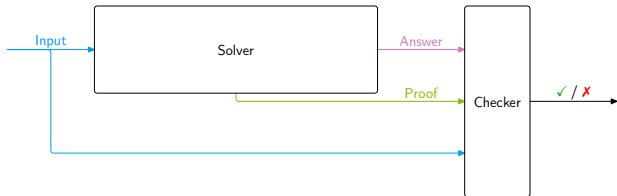


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Clear conflict expressivity vs. simplicity!

Proof Logging Desiderata



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Clear conflict expressivity vs. simplicity!

Asking for both perhaps a little bit too good to be true?

Some Previous Proof Logging Work

Boolean satisfiability (SAT) solving

- Well established since over decade with several proof formats such as [DRAT](#) [HHW13a, HHW13b, WHH14], [GRIT](#) [CMS17], [LRAT](#) [CHH⁺17], ...
- But no efficient support for most advanced techniques such as
 - Gaussian elimination
 - symmetry breaking
- And only for decision problems, not optimisation (MaxSAT)

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Mixed integer linear programming

- Work on proof format [VIPR](#) [CGS17, EG23]
- But only for exact solving and without support for advanced techniques

Take-Away Message from These Lectures

Proof logging for sophisticated combinatorial solvers is possible!



Take-Away Message from These Lectures



Proof logging for sophisticated combinatorial solvers is possible!

- With **single, unified method!**
- Producing proofs in simple format calculating with **0–1 integer linear inequalities** (a.k.a. **pseudo-Boolean constraints**)
- Implemented in **VeriPB** (<https://VeriPB.org>)

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- 8 Offers a **framework for AI coding agents** by requiring that code should generate correct proofs

Design Principles for Proof Logging

Proof logging implementation

- Don't change solver
- Just add proof logging print statements (plus some book-keeping) to solver code

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- Proof logging overhead small constant fraction of running time ($\lesssim 10\%$)
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Proof system

- Keep language simple — no XOR constraints, CP propagators, symmetry groups, ...
- But reason efficiently about such notions using power of proof system
- Combine proof logging with formally verified proof checker

But Let Us Start from the Beginning...

Review of some basic concepts:

- SATISFIABILITY (SAT) problem
- Unit propagation
- DPLL and CDCL algorithms
- Resolution proof system

The SATISFIABILITY (SAT) Problem

- **Variable** x : takes value **true** (=1) or **false** (=0)
- **Literal** ℓ : variable x or its negation $\bar{x}$
- **Clause** $C = \ell_1 \vee \dots \vee \ell_k$: disjunction of literals
(Consider as sets, so no repetitions and order irrelevant)
- **Conjunctive normal form (CNF) formula** $F = C_1 \wedge \dots \wedge C_m$: conjunction of clauses

The SAT Problem

Given a CNF formula F , is it satisfiable?

For instance, what about:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge \\ (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

Proofs for SAT

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For unsatisfiability: a sequence of clauses (CNF constraints)

- Each clause follows “obviously” from everything we know so far
- Final clause is empty, meaning contradiction (written $\perp$)
- Means original formula must be inconsistent

What Is Obvious? Unit Propagation

Unit Propagation

Clause C **unit propagates** ℓ under partial assignment ρ if ρ falsifies all literals in C except ℓ

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Proof checker should know how to unit propagate until saturation

Davis-Putnam-Logemann-Loveland (DPLL)

DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

“Proof trace”: when backtracking, write negation of guesses made

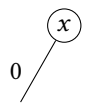
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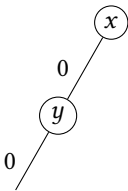


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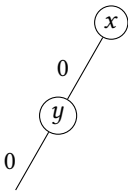


Davis-Putnam-Logemann-Loveland (DPLL)

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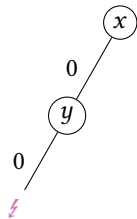
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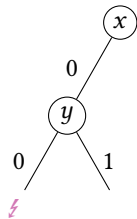
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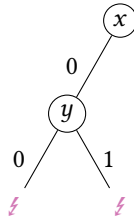
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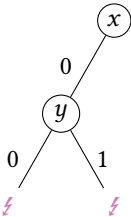
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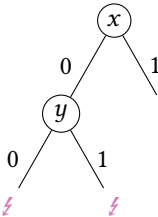
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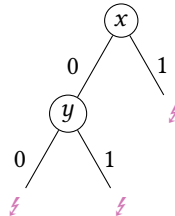
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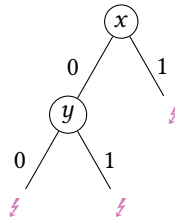
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5 $\perp$



Reverse Unit Propagation (RUP)

To make this into a proof, need backtrack clauses to be easily verifiable

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Fact

Backtrack clauses from DPLL solver generate a proof consisting of RUP clauses

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Run CDCL [BS97, MS99, MMZ⁺01] on our favourite CNF formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

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Free choice to assign value to variable

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Forced choice to avoid falsifying clause

Given $p = 0$, clause $p \vee \bar{u}$ forces $u = 0$

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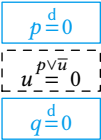
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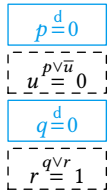
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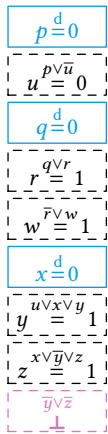
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$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{?}{=}$$

decision
level 1

decision
level 2

decision
level 3

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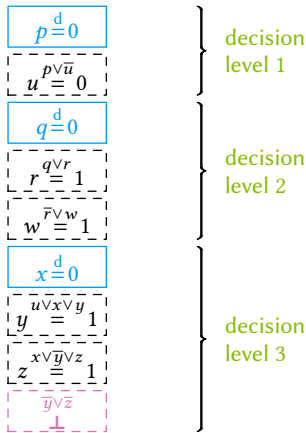
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Conflict Analysis

Time to analyse this conflict and learn from it!

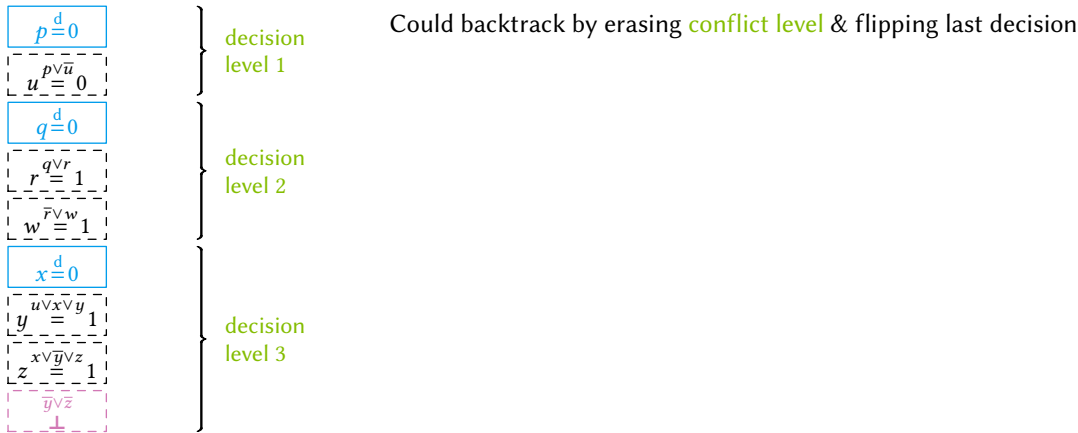
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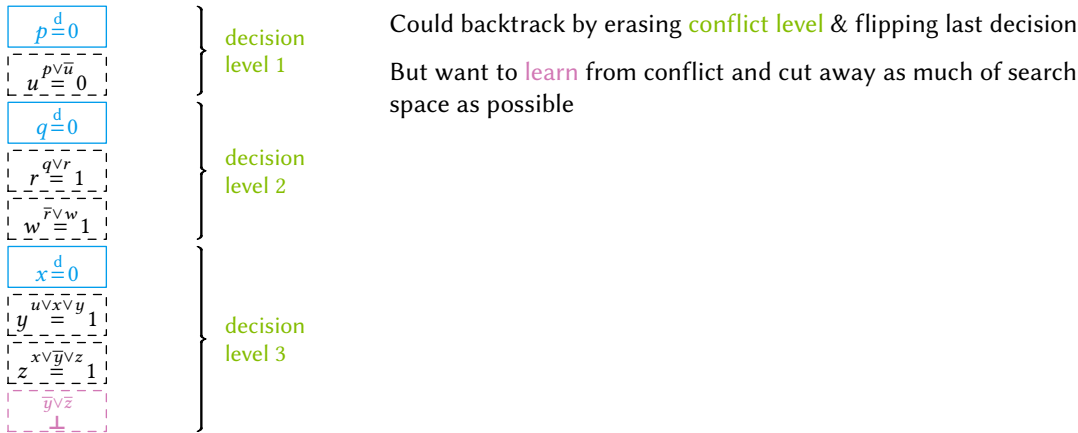
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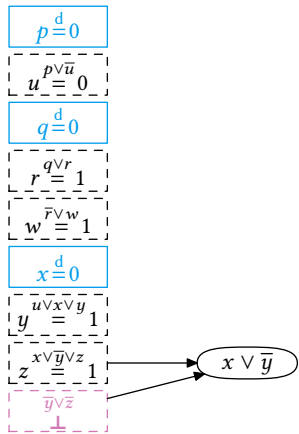
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Could backtrack by erasing **conflict level** & flipping last decision

But want to **learn** from conflict and cut away as much of search space as possible

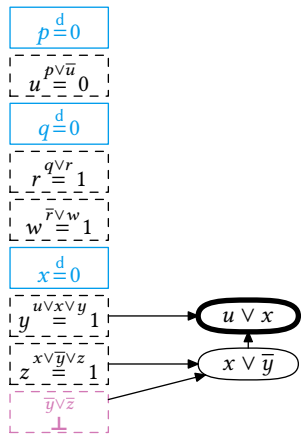
Case analysis over z for last two clauses:

- $x \vee \bar{y} \vee z$ wants $z = 1$
- $\bar{y} \vee \bar{z}$ wants $z = 0$
- **Resolve** clauses by merging them & removing z — must satisfy $x \vee \bar{y}$

Conflict Analysis

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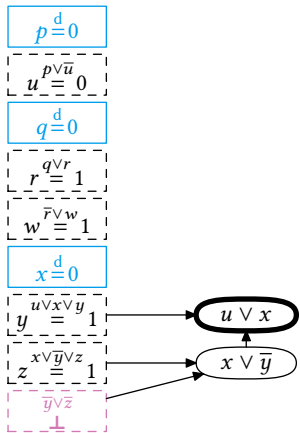
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Repeat until **UIP clause** with only 1 variable at conflict level after last decision — **learn** and **backjump**

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

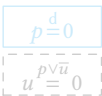
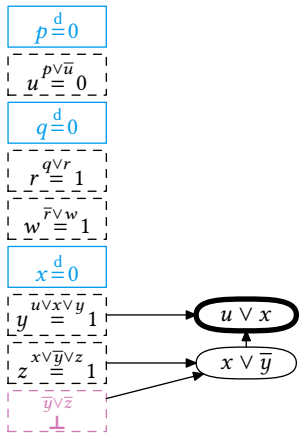
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Complete Example of CDCL Execution

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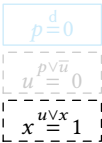
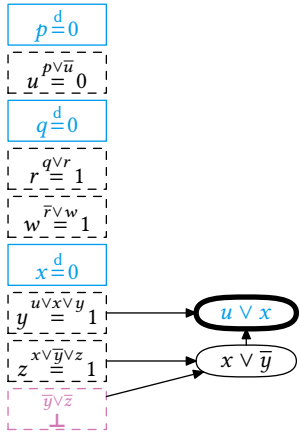


Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

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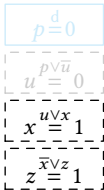
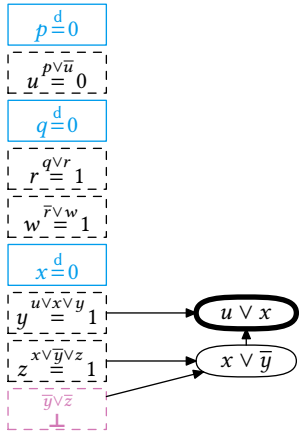
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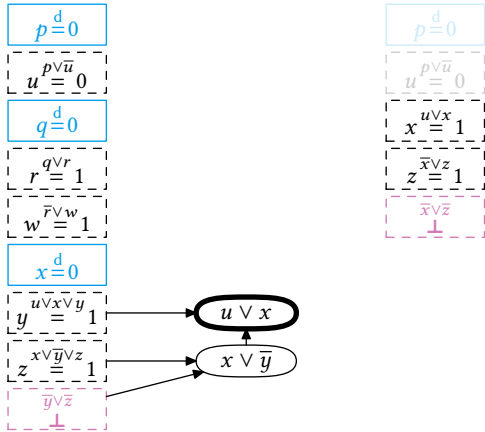
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Then continue as before...

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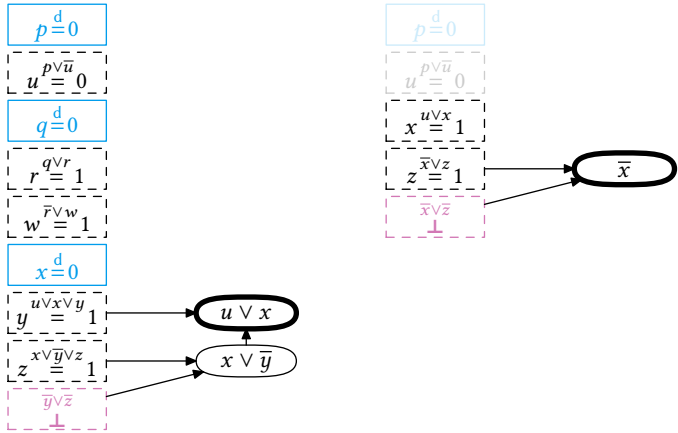
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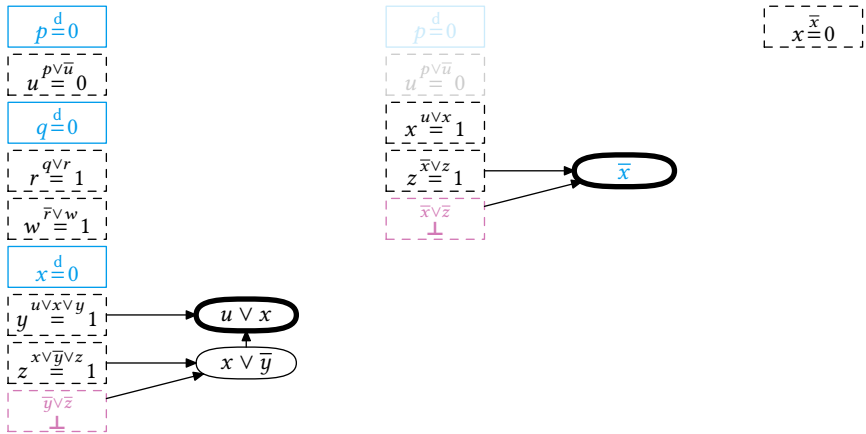
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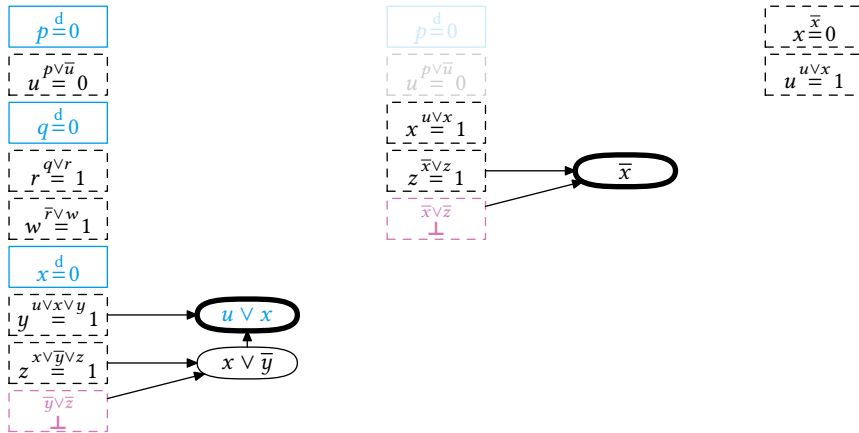
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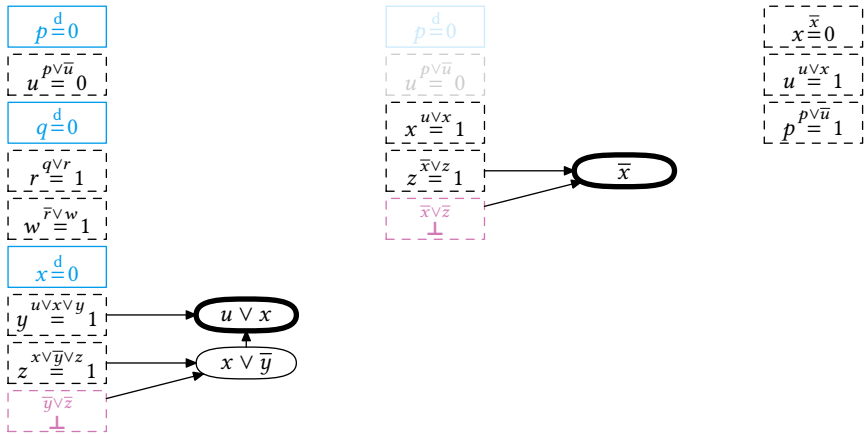
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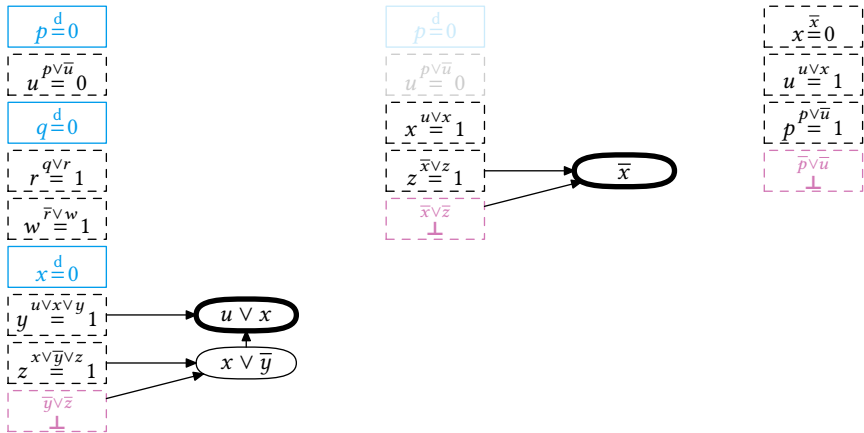
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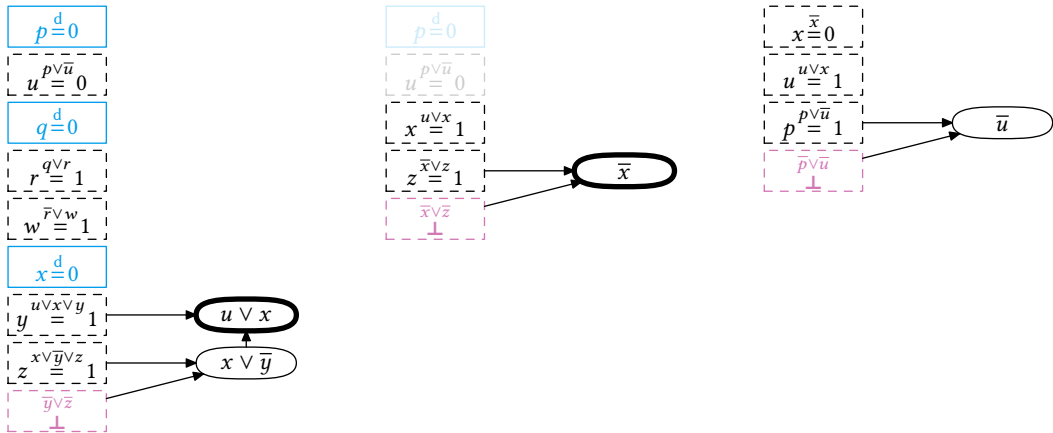
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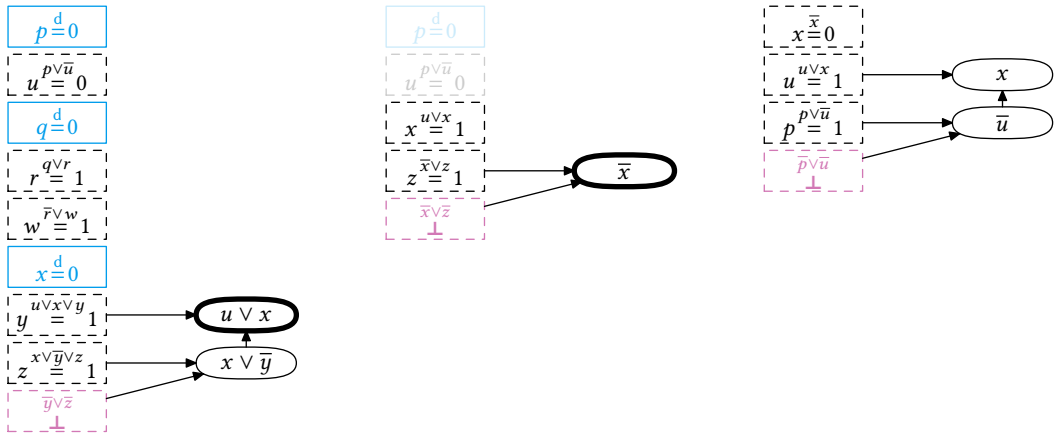
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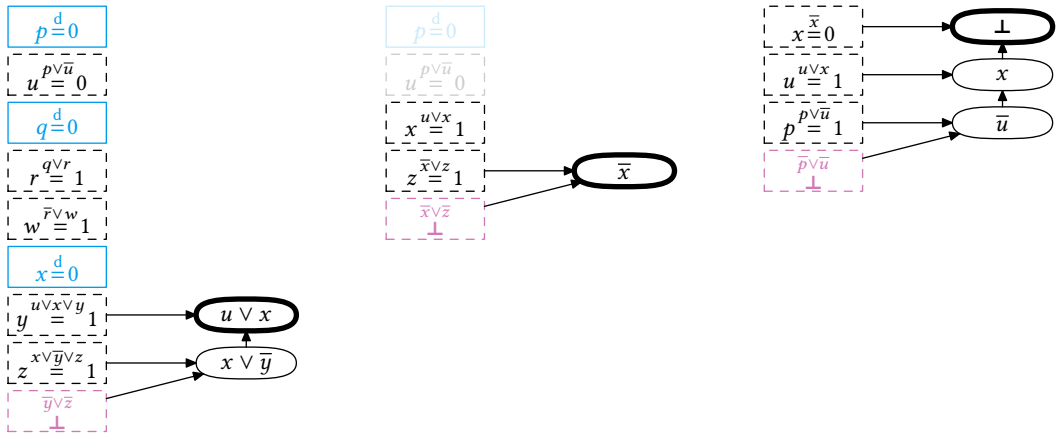
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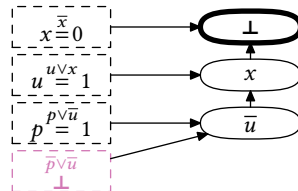
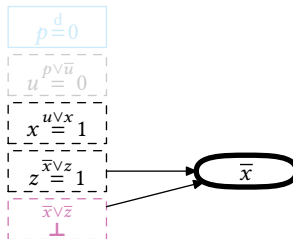
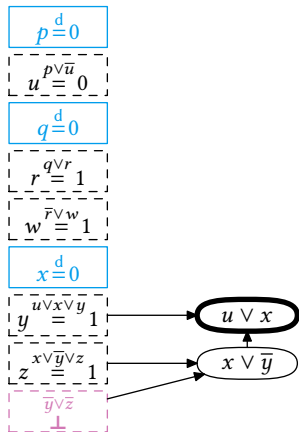
(*) Ignores pre- and inprocessing, but we will get there...

Resolution Proofs from CDCL Executions

Obtain resolution proof...

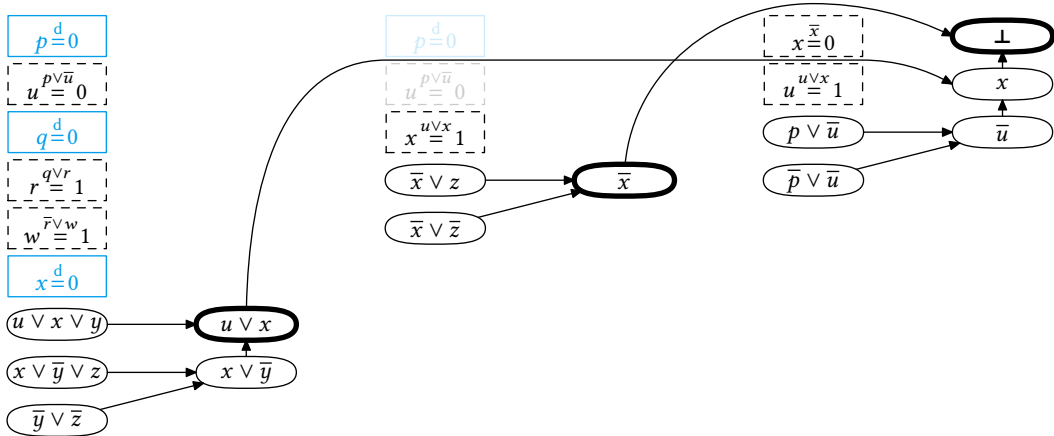
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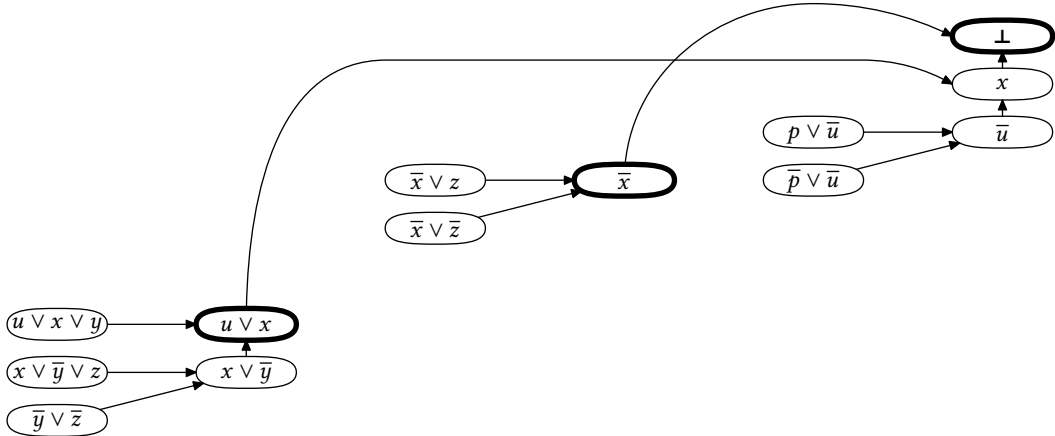
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More Ingredients in Proof Logging for SAT

Fact

RUP proofs can be viewed as shorthand for resolution proofs

See proof complexity and SAT solving survey [BN21] for more on this

But RUP and resolution are not enough for preprocessing, inprocessing, and some other kinds of reasoning

Extension Variables, Part 1

Suppose we want a **reified variable** r encoding

$$r \Leftrightarrow (x \wedge y)$$

Extended resolution [Tse68]

Resolution rule plus **extension rule** introducing clauses

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Fact

Extended resolution (RUP + definition of new variables) is essentially equivalent to the DRAT proof logging system most commonly used for SAT solving [KRH18]

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Practical limitations of current SAT proof logging technology:

- Difficulties dealing with stronger reasoning efficiently (even for SAT solving)
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- And more...

Pseudo-Boolean Constraints

0–1 integer linear inequalities or (linear) pseudo-Boolean constraints:

$$\sum_i a_i \ell_i \geq A$$

- $a_i, A \in \mathbb{Z}$
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Sometimes convenient to use **normalised form** [Bar95] with **all a_i, A non-negative** (without loss of generality)

Some Types of Pseudo-Boolean Constraints

1 Clauses

$$x_1 \vee \bar{x}_2 \vee x_3 \quad \Leftrightarrow \quad x_1 + \bar{x}_2 + x_3 \geq 1$$

2 Cardinality constraints

$$x_1 + x_2 + x_3 + x_4 \geq 2$$

3 General pseudo-Boolean constraints

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms

From the input

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$$\overline{\ell_i \geq 0}$$

Addition

$$\frac{\sum_i a_i \ell_i \geq A \quad \sum_i b_i \ell_i \geq B}{\sum_i (a_i + b_i) \ell_i \geq A + B}$$

Multiplication for any $c \in \mathbb{N}^+$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i c a_i \ell_i \geq c A}$$

Division for any $c \in \mathbb{N}^+$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \lceil \frac{a_i}{c} \rceil \ell_i \geq \lceil \frac{A}{c} \rceil}$$

Saturation for any $c \in \mathbb{N}^+$
(assumes normalised form)

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \min(a_i, A) \cdot \ell_i \geq A}$$

Cutting Planes Toy Example

$$w + 2x + y \geq 2$$

Cutting Planes Toy Example

Multiply by 2

$$\frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$$

Cutting Planes Toy Example

Multiply by 2

$$\frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$$

$$w + 2x + 4y + 2z \geq 5$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 & w + 2x + y \geq 2 & \\
 \text{Multiply by 2} & \frac{2w + 4x + 2y \geq 4}{} & w + 2x + 4y + 2z \geq 5 \\
 \text{Add} & \frac{2w + 4x + 2y \geq 4}{3w + 6x + 6y + 2z \geq 9} &
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
 \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \bar{z} \geq 0
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
 \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \\
 & & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} \text{ Multiply by 2}
 \end{array}$$

Cutting Planes Toy Example

Multiply by 2

$$\begin{array}{r} w + 2x + y \geq 2 \\ \hline 2w + 4x + 2y \geq 4 \end{array}$$

Add

$$\begin{array}{r} 2w + 4x + 2y \geq 4 \\ w + 2x + 4y + 2z \geq 5 \\ \hline 3w + 6x + 6y + 2z \geq 9 \end{array}$$

Add

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \\ 2\bar{z} \geq 0 \\ \hline 3w + 6x + 6y + 2z + 2\bar{z} \geq 9 \end{array}$$

Multiply by 2

$$\begin{array}{r} \bar{z} \geq 0 \\ \hline 2\bar{z} \geq 0 \end{array}$$

Cutting Planes Toy Example

Multiply by 2

$$\begin{array}{r} w + 2x + y \geq 2 \\ \hline 2w + 4x + 2y \geq 4 \end{array}$$

Add

$$\begin{array}{r} 2w + 4x + 2y \geq 4 \\ w + 2x + 4y + 2z \geq 5 \\ \hline 3w + 6x + 6y + 2z \geq 9 \end{array}$$

Add

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \\ \hline 3w + 6x + 6y + 2 \geq 9 \end{array}$$

Multiply by 2

$$\begin{array}{r} \bar{z} \geq 0 \\ \hline 2\bar{z} \geq 0 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
 \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \\
 & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} & \text{Multiply by 2} \\
 \text{Add} & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} & \geq 7
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
 \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} \quad \text{Multiply by 2} \\
 \text{Add} & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} & \geq 7 \\
 \text{Divide by 3} & \frac{3w + 6x + 6y}{w + 2x + 2y} & \geq 2\frac{1}{3}
 \end{array}$$

Cutting Planes Toy Example

$$\begin{array}{rcl}
 \text{Multiply by 2} & \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} & \\
 \text{Add} & \frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9} & \frac{\bar{z} \geq 0}{2\bar{z} \geq 0} \quad \text{Multiply by 2} \\
 \text{Add} & \frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} & \geq 7 \\
 \text{Divide by 3} & \frac{3w + 6x + 6y}{w + 2x + 2y} & \geq 3
 \end{array}$$

Cutting Planes Toy Example

Multiply by 2

$$\frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$$

Add

$$\frac{w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9}$$

Add

$$\frac{\bar{z} \geq 0}{2\bar{z} \geq 0}$$

Multiply by 2

$$\frac{2\bar{z} \geq 0}{2\bar{z} \geq 0}$$

$$\frac{3w + 6x + 6y + 2z \geq 9}{3w + 6x + 6y} \geq 7$$

Divide by 3

$$\frac{3w + 6x + 6y}{w + 2x + 2y} \geq 3$$

Naming constraints by integers and literal axioms by the literal involved (with ~ for negation) as

Constraint 1

$\doteq 2x + y + w \geq 2$

Constraint 2

$\doteq 2x + 4y + 2z + w \geq 5$

~z

$\doteq \bar{z} \geq 0$

Cutting Planes Toy Example

Multiply by 2

$$\begin{array}{r} w + 2x + y \geq 2 \\ \hline 2w + 4x + 2y \geq 4 \end{array}$$

Add

$$\begin{array}{r} 2w + 4x + 2y \geq 4 \\ w + 2x + 4y + 2z \geq 5 \\ \hline 3w + 6x + 6y + 2z \geq 9 \end{array}$$

Add

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \\ \bar{z} \geq 0 \\ \hline 2\bar{z} \geq 0 \end{array}$$

Multiply by 2

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \\ 2\bar{z} \geq 0 \\ \hline 3w + 6x + 6y \geq 7 \end{array}$$

Divide by 3

$$\begin{array}{r} 3w + 6x + 6y \geq 7 \\ \hline w + 2x + 2y \geq 3 \end{array}$$

Naming constraints by integers and literal axioms by the literal involved (with ~ for negation) as

Constraint 1

$\doteq 2x + y + w \geq 2$

Constraint 2

$\doteq 2x + 4y + 2z + w \geq 5$

~z

$\doteq \bar{z} \geq 0$

such a calculation is written in the proof log in reverse Polish notation as*

pol

1

2

*

2

+

~z

2

*

+

3

d

;

Cutting Planes Toy Example

Multiply by 2

$$\begin{array}{r} w + 2x + y \geq 2 \\ \hline 2w + 4x + 2y \geq 4 \end{array}$$

Add

$$\begin{array}{r} 2w + 4x + 2y \geq 4 \qquad w + 2x + 4y + 2z \geq 5 \\ \hline 3w + 6x + 6y + 2z \geq 9 \end{array}$$

Add

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \qquad \bar{z} \geq 0 \\ \hline 3w + 6x + 6y \qquad \qquad \geq 7 \end{array}$$

Multiply by 2

$$\begin{array}{r} 3w + 6x + 6y \qquad \qquad \geq 7 \\ \hline 2\bar{z} \geq 0 \end{array}$$

Add

$$\begin{array}{r} 3w + 6x + 6y \qquad \qquad \geq 7 \\ \hline 3w + 6x + 6y + 2z \geq 9 \end{array}$$

Divide by 3

$$\begin{array}{r} 3w + 6x + 6y + 2z \geq 9 \\ \hline w + 2x + 2y + \frac{2}{3}z \geq 3 \end{array}$$

Naming constraints by integers and literal axioms by the literal involved (with ~ for negation) as

Constraint 1

$\doteq 2x + y + w \geq 2$

Constraint 2

$\doteq 2x + 4y + 2z + w \geq 5$

~z

$\doteq \bar{z} \geq 0$

such a calculation is written in the proof log in reverse Polish notation as*

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d

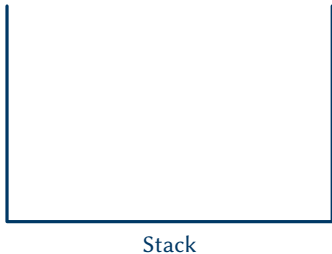
;

(*) Except VERIPB variables have to be at least two characters long, but we abuse syntax slightly here for simplicity

Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

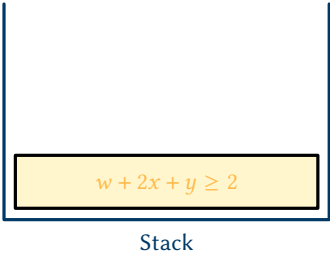
Read pol line from left to right, starting with empty stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

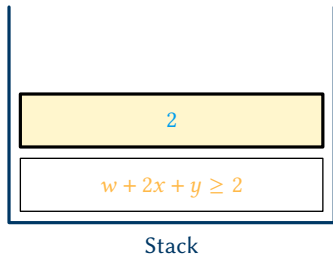
Read 1: push constraint 1 onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

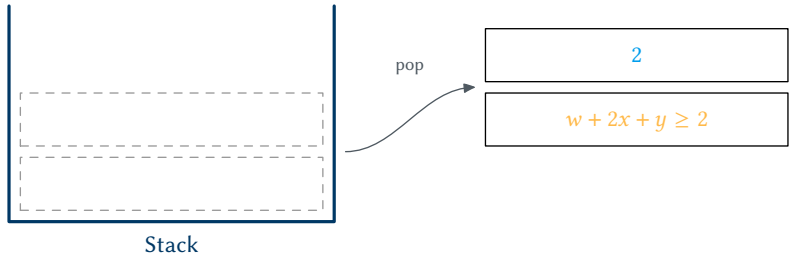
Read 2: push constant 2 onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Read *: pop two elements from stack



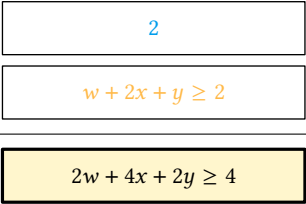
Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Multiply constraint by 2



Stack

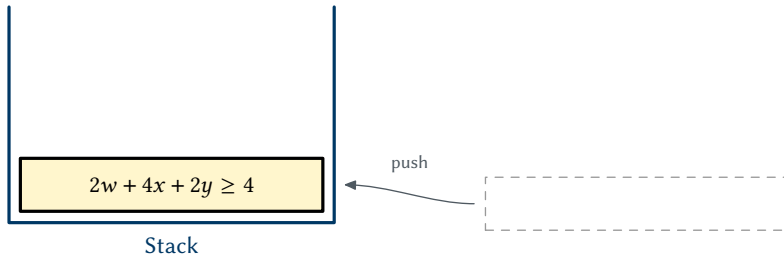


Multiply by 2

Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

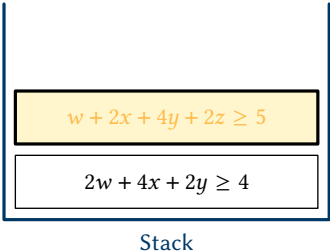
Push result onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

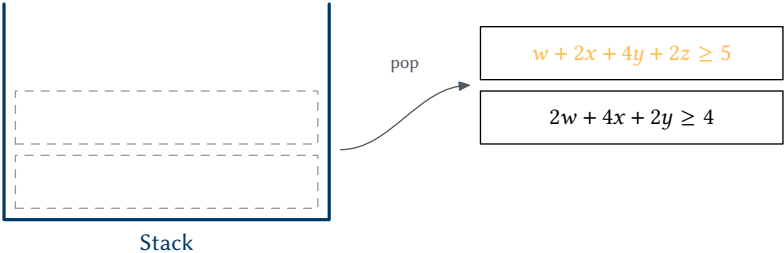
Read 2: push constraint 2 onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Read +: pop two elements from stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Add constraints



Stack

$w + 2x + 4y + 2z \geq 5$

$2w + 4x + 2y \geq 4$

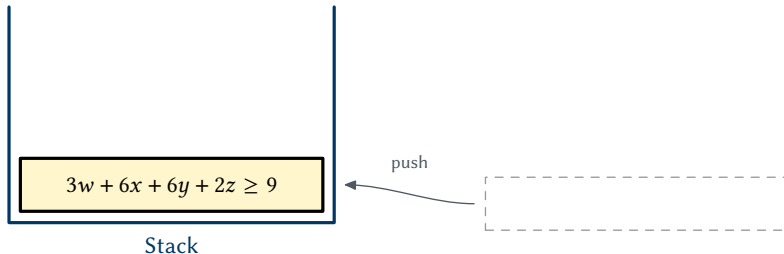
$3w + 6x + 6y + 2z \geq 9$

Add

Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

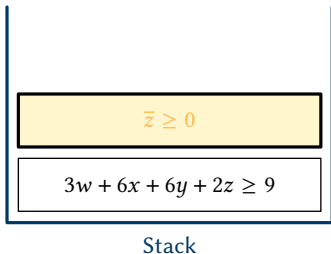
Push result onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + $\bar{z}$ 2 * + 3 d ;

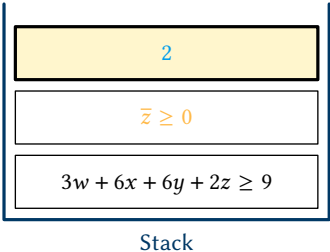
Read $\bar{z}$: push literal axiom $\bar{z} \geq 0$ onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

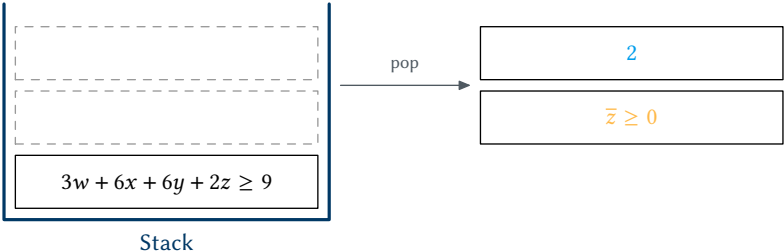
Read 2: push constant 2 onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

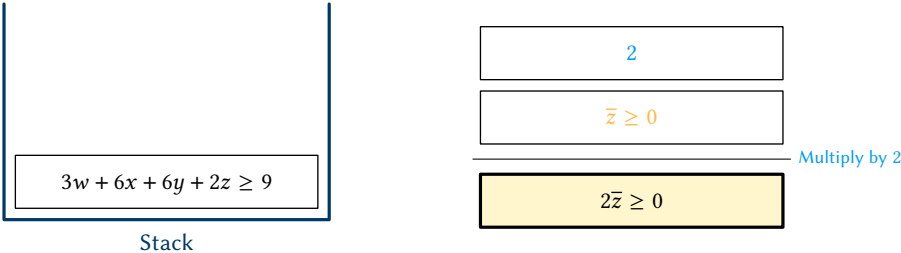
Read *: pop two elements from stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

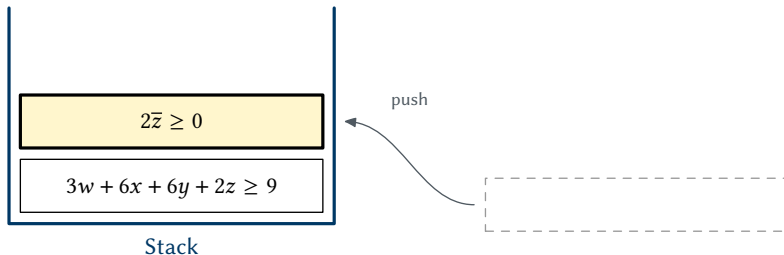
Multiply constraint by 2



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

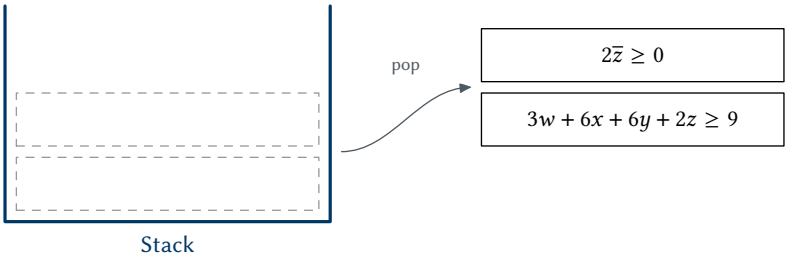
Push result onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Read +: pop two elements from stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

Add constraints



Stack

$$2\bar{z} \geq 0$$

$$3w + 6x + 6y + 2z \geq 9$$

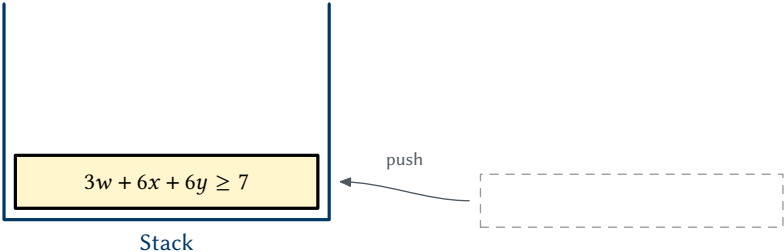
$$3w + 6x + 6y \geq 7$$

Add

Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

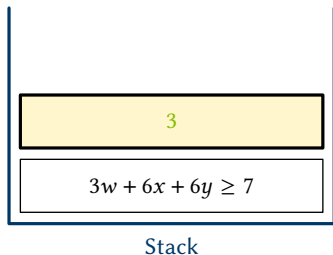
Push result onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

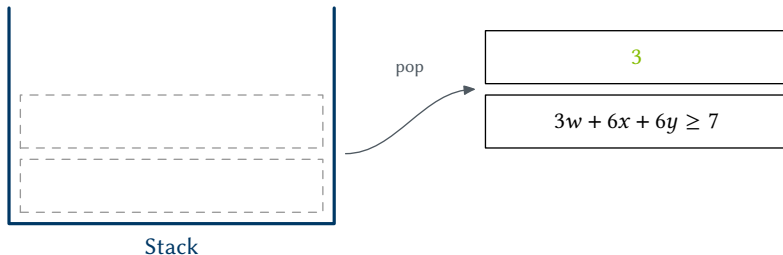
Read 3: push constant 3 onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 **d** ;

Read d: pop two elements from stack



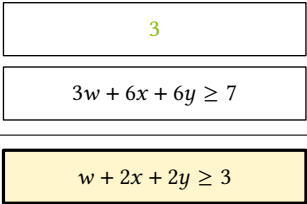
Evaluating a Reverse Polish Notation Line Using a Stack

```
pol 1 2 * 2 + ~z 2 * + 3 d ;
```

Divide constraint by 3



Stack

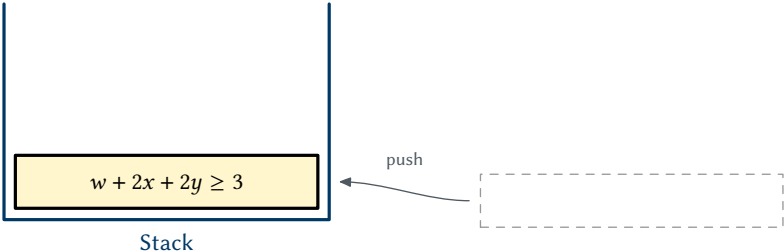


Divide by 3

Evaluating a Reverse Polish Notation Line Using a Stack

```
pol 1 2 * 2 + ~z 2 * + 3 d ;
```

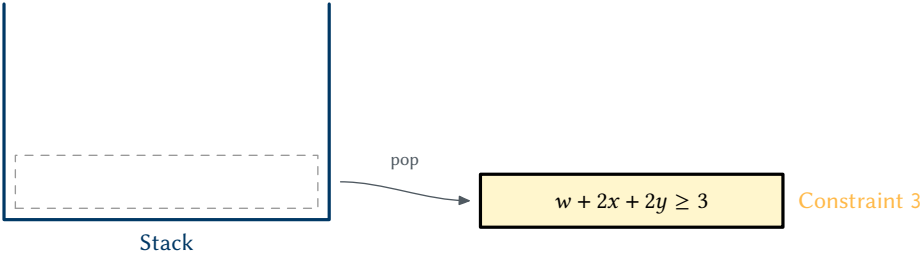
Push result onto stack



Evaluating a Reverse Polish Notation Line Using a Stack

pol 1 2 * 2 + ~z 2 * + 3 d ;

End of line: pop final result and add it to proof with next available constraint ID (here 3, say)



Resolution and Cutting Planes

To simulate resolution step such as

$$\frac{\bar{y} \vee \bar{z} \quad x \vee \bar{y} \vee z}{x \vee \bar{y}}$$

we can perform the cutting planes steps

$$\begin{array}{l} \text{Add} \quad \frac{\bar{y} + \bar{z} \geq 1 \quad x + \bar{y} + z \geq 1}{x + 2\bar{y} \geq 1} \\ \text{Divide by 2} \quad \frac{x + 2\bar{y} \geq 1}{x + \bar{y} \geq 1} \end{array}$$

Resolution and Cutting Planes

To simulate resolution step such as

$$\frac{\overline{y} \vee \overline{z} \qquad x \vee \overline{y} \vee z}{x \vee \overline{y}}$$

we can perform the cutting planes steps

$$\begin{array}{l} \text{Add} \quad \frac{\overline{y} + \overline{z} \geq 1 \qquad x + \overline{y} + z \geq 1}{x + 2\overline{y} \geq 1} \\ \text{Divide by 2} \quad \frac{x + 2\overline{y} \geq 1}{x + \overline{y} \geq 1} \end{array}$$

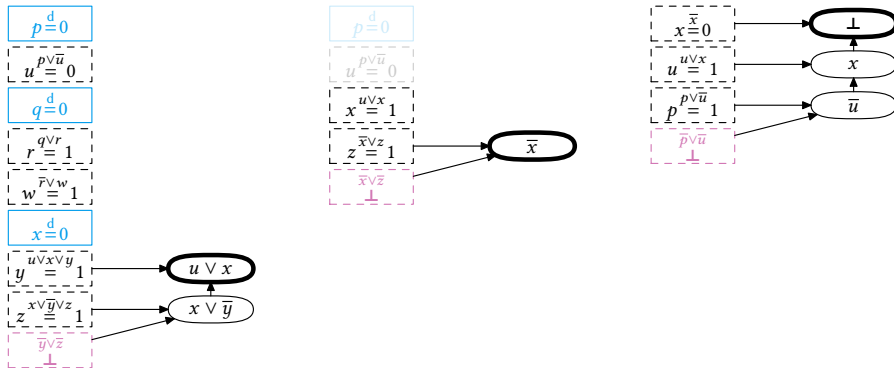
Given that the premises are clauses 7 and 5 in our example CNF formula, using references

$$\begin{array}{l} \text{Constraint 7} \doteq \overline{y} + \overline{z} \geq 1 \\ \text{Constraint 5} \doteq x + \overline{y} + z \geq 1 \end{array}$$

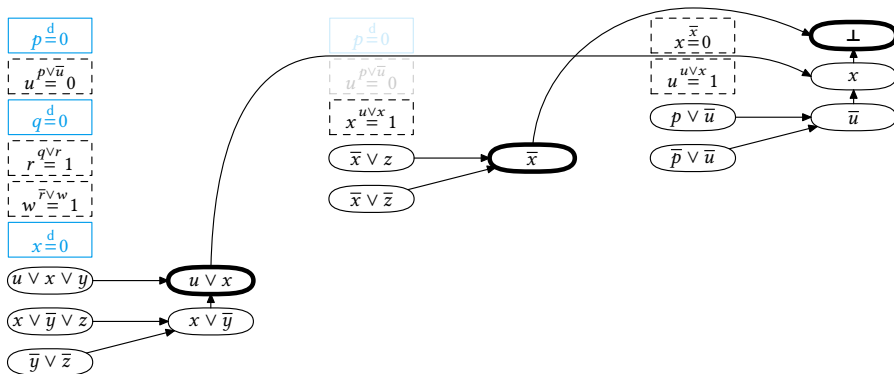
we can write this in the proof log as

$$\text{pol } 7 \ 5 \ + \ 2 \ d \ ;$$

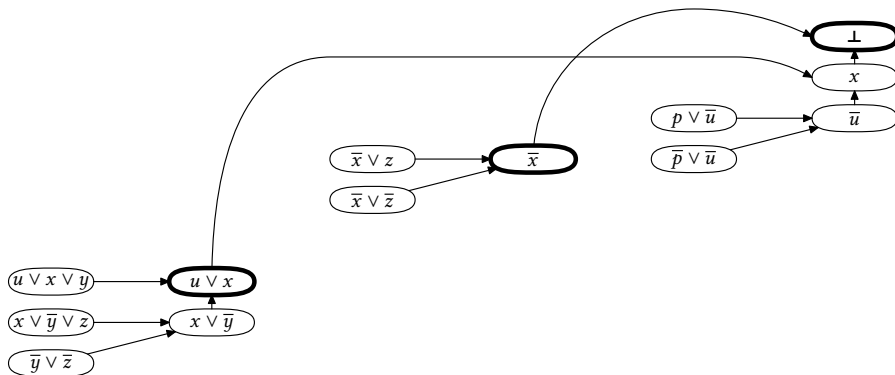
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



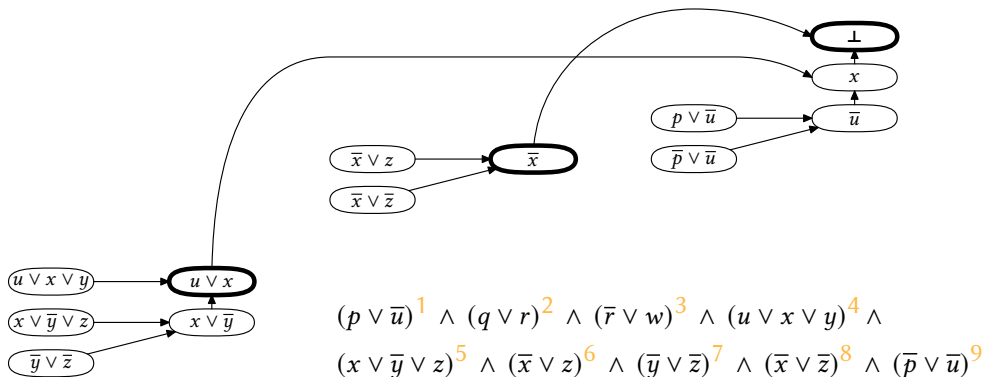
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



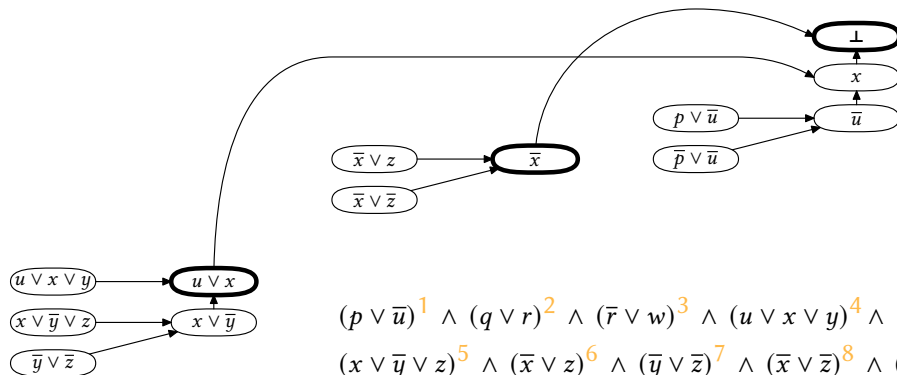
Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



Pseudo-Boolean Proof Logging for Example CDCL Conflict Analyses



$$(p \vee \bar{u})^1 \wedge (q \vee r)^2 \wedge (\bar{r} \vee w)^3 \wedge (u \vee x \vee y)^4 \wedge (x \vee \bar{y} \vee z)^5 \wedge (\bar{x} \vee z)^6 \wedge (\bar{y} \vee \bar{z})^7 \wedge (\bar{x} \vee \bar{z})^8 \wedge (\bar{p} \vee \bar{u})^9$$

$$\text{pol } 7 \ 5 \ + \ 2 \ d \ 4 \ + \ 2 \ d \ ;$$

$$\text{pol } 8 \ 6 \ + \ 2 \ d \ ;$$

$$\text{pol } 9 \ 1 \ + \ 2 \ d \ 10 \ + \ 2 \ d \ 11 \ + \ 2 \ d \ ;$$

$$\rightsquigarrow \text{Constraint } 10 \doteq u + x \geq 1$$

$$\rightsquigarrow \text{Constraint } 11 \doteq \bar{x} \geq 1$$

$$\rightsquigarrow \text{Constraint } 12 \doteq 0 \geq 1 \quad \text{⚡}$$

RUP Revisited

Can define (reverse) unit propagation in a pseudo-Boolean setting [EGMN20]

Constraint C propagates variable x if setting x to “wrong value” would make C unsatisfiable

E.g., if x_5 is false,

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

propagates $\bar{x}_4$ (since other coefficients do not add up to 7)

RUP Revisited

Can define (reverse) unit propagation in a pseudo-Boolean setting [EGMN20]

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E.g., if x_5 is false,

$$x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

propagates $\bar{x}_4$ (since other coefficients do not add up to 7)

Risk for confusion!

- Constraint programming people might call this (reverse) integer bounds consistency
 - Does the same thing if we're working with clauses
 - More interesting for general pseudo-Boolean constraints
- SAT people beware: constraints can propagate multiple times and multiple variables

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$

Assuming normalised form:

$$\text{slack}(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Propagation, Conflict, and Slack

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Assuming normalised form:

$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

ρ	$slack(C; \rho)$	comment

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$
Assuming normalised form:

$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

ρ	$slack(C; \rho)$	comment
$\{\}$	8	

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$
Assuming normalised form:

$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

ρ	$slack(C; \rho)$	comment
$\{\}$	8	
$\{\bar{x}_5\}$	3	propagates $\bar{x}_4$ (coefficient > slack)

Propagation, Conflict, and Slack

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Assuming normalised form:

$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

ρ	$slack(C; \rho)$	comment
$\{\}$	8	
$\{\bar{x}_5\}$	3	propagates $\bar{x}_4$ (coefficient > slack)
$\{\bar{x}_5, \bar{x}_4\}$	3	propagation doesn't change slack

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$
Assuming normalised form:

$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

ρ	$slack(C; \rho)$	comment
$\{\}$	8	
$\{\bar{x}_5\}$	3	propagates $\bar{x}_4$ (coefficient > slack)
$\{\bar{x}_5, \bar{x}_4\}$	3	propagation doesn't change slack
$\{\bar{x}_5, \bar{x}_4, \bar{x}_3, x_2\}$	-2	conflict (slack < 0)

Propagation, Conflict, and Slack

Slack measures how far assignment ρ is from falsifying $\sum_i a_i \ell_i \geq A$
Assuming normalised form:

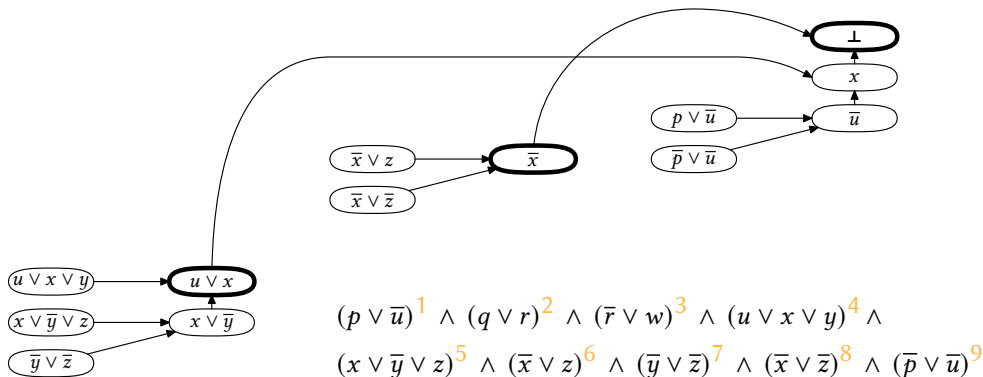
$$slack(\sum_i a_i \ell_i \geq A; \rho) = \sum_{i: \rho(\ell_i) \neq 0} a_i - A$$

Consider $C \doteq x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$

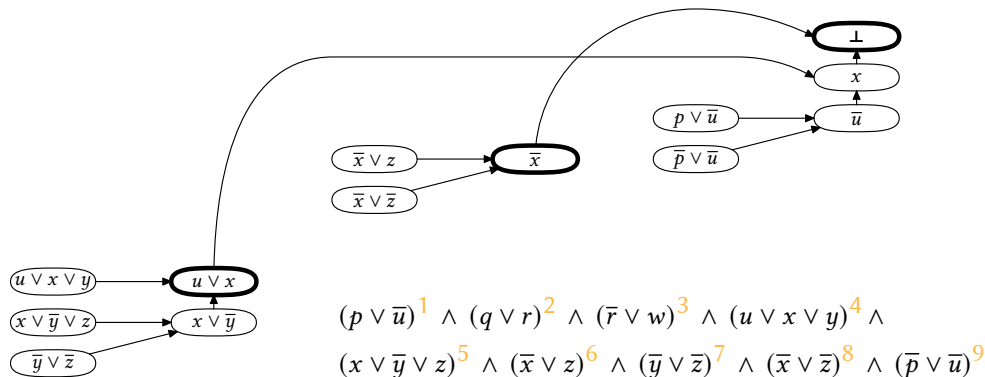
ρ	$slack(C; \rho)$	comment
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Note: constraint can be conflicting though not all variables assigned

Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



Pseudo-Boolean Proof Logging for Example CDCL Execution with RUP



rup 1 u 1 x >= 1 ;
 rup 1 ~x >= 1 ;
 rup >= 1 ;

$\rightsquigarrow$ Constraint 10 $\doteq u + x \geq 1$

$\rightsquigarrow$ Constraint 11 $\doteq \bar{x} \geq 1$

$\rightsquigarrow$ Constraint 12 $\doteq 0 \geq 1$ ⚡

Extension Variables, Part 2

Suppose we want new, fresh **reified variable** r encoding

$$r \Leftrightarrow (3x + 2y + z + w \geq 3)$$

This time, introduce constraints

$$3\bar{r} + 3x + 2y + z + w \geq 3 \qquad 5r + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

Again, needs support from the proof system in the form of **strengthening rules**

Proof Logs for “Cutting Planes with Strengthening”

For satisfiable instances: just specify a satisfying assignment

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- Either implicitly, by RUP...
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(*) Not actually implemented this way — more details in a few slides ...

Optimisation and Enumeration Problems

Optimisation

- Define an objective function $f = \sum_i w_i \ell_i + k$, $w_i, k \in \mathbb{Z}$, to minimise subject to the constraints in the formula
- To maximise, negate objective
- Log solution α with `sol1` rule $\Rightarrow$ objective function value $UB = \sum_i w_i \alpha(\ell_i) + k$
- Add **objective-improving constraint** $f \leq UB - 1$
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Enumeration

- When a solution is found, can log it with solx rule
- Introduces a new constraint saying “not this solution”
- So the proof semantics is “infeasible, except for all the solutions I told you about”

Deleting Constraints

- In practice, important to erase constraints to save memory and time during verification
- Unsatisfiability proofs: fairly straightforward to deal with from point of view of proof logging
- Optimisation and enumeration proofs: significantly more delicate
- We will mostly ignore deletions during these lectures for simplicity and clarity

Pseudo-Boolean Proof Logging — How and Why?

If problem is (special case of) 0–1 integer linear program

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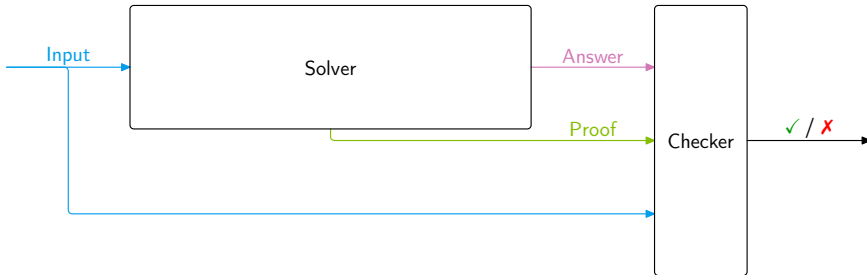
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$$7\bar{r} + x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

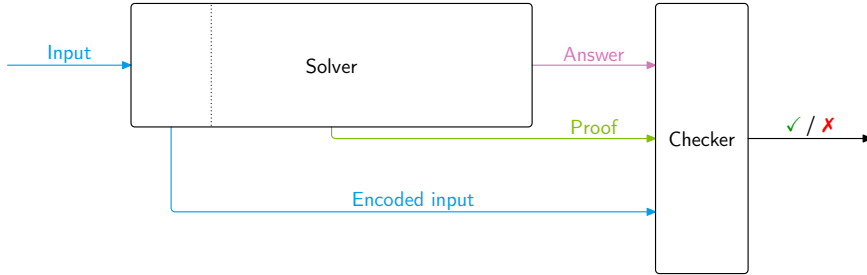
$$r \Leftarrow x_1 + 2\bar{x}_2 + 3x_3 + 4\bar{x}_4 + 5x_5 \geq 7$$

$$9r + \bar{x}_1 + 2x_2 + 3\bar{x}_3 + 4x_4 + 5\bar{x}_5 \geq 9$$

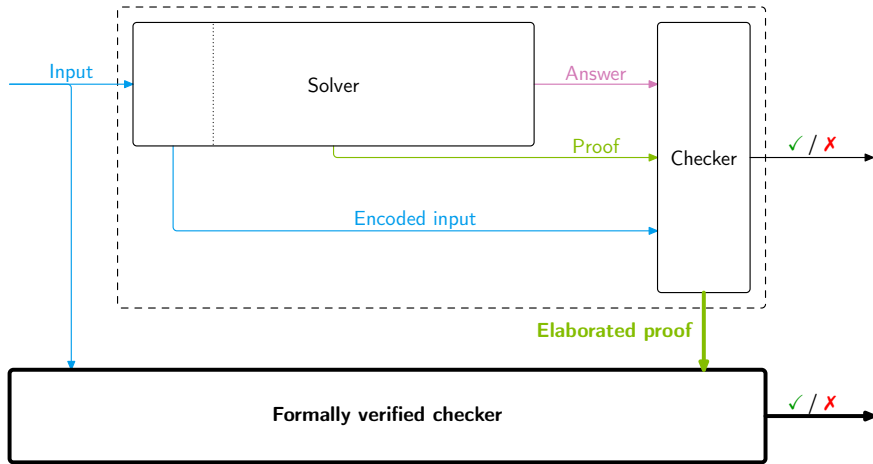
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Wish to allow **without-loss-of-generality** arguments that can derive non-implied constraints

“Redundant” Constraints

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But how to certify redundant constraints?

Redundance-Based Strengthening

Redundance-based strengthening [BT19, GN21] (extending RAT rule of SAT proof logging)

C is redundant with respect to F if and only if there is a **substitution** ω (mapping variables to truth values or literals), called a **witness**, for which

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- 2 by explicitly presented derivation

Deriving the reification $r \Leftrightarrow (3x + 2y + z + w \geq 3)$

Want to derive

$$3\bar{r} + 3x + 2y + z + w \geq 3 \quad 5r + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

encoding equivalence $r \Leftrightarrow (3x + 2y + z + w \geq 3)$ for fresh variable r

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Can we make this into a formal proof?

Deriving $r \Leftrightarrow (3x + 2y + z + w \geq 3)$ Using the Redundance Rule

Make formal derivation of

$$3\bar{r} + 3x + 2y + z + w \geq 3 \quad 5r + 3\bar{x} + 2\bar{y} + \bar{z} + \bar{w} \geq 5$$

using condition $F \cup \{\neg C\} \models (F \cup \{C\}) \upharpoonright_{\omega}$

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VERIPB code:

```
red 3 ~r 3 x 2 y 1 z 1 w >= 3 : r -> 0 ;
```

```
red 5 r 3 ~x 2 ~y 1 ~z 1 ~w >= 5 : r -> 1 ;
```

Redundance and Dominance Rules for Optimisation

Redundance-based strengthening, optimisation version

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Dominance-based strengthening (simplified)

Add constraint C to formula F if exists witness substitution ω s.t.

$$F \cup \{\neg C\} \models F \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} < f\}$$

Redundance and Dominance Rules for Optimisation

Redundance-based strengthening, optimisation version

Add constraint C to formula F if exists witness substitution ω s.t.

$$F \cup \{\neg C\} \models (F \cup \{C\}) \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} \leq f\}$$

Can be more aggressive if witness ω **strictly improves** solution

Dominance-based strengthening (simplified)

Add constraint C to formula F if exists witness substitution ω s.t.

$$F \cup \{\neg C\} \models F \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} < f\}$$

- Applying ω should **strictly decrease** f
- If so, don't need to show that $C \upharpoonright_{\omega}$ is implied!

Soundness of Dominance Rule

Dominance-based strengthening (simplified)

Add constraint C to formula F if exists witness substitution ω s.t.

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Why is this sound?

Soundness of Dominance Rule

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- 2 Then $\alpha \circ \omega$ satisfies F and $f(\alpha \circ \omega) < f(\alpha)$

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- 7 ...
- 8 Can't go on forever, so finally reach α' satisfying both F and C

Strength of Dominance Rule

Dominance-based strengthening (stronger, still simplified)

If $C_1, C_2, \dots, C_{m-1}$ have been derived from F (maybe using dominance), then can derive C_m if exists witness substitution ω s.t.

$$F \cup \{C_1, \dots, C_{m-1}\} \cup \{\neg C_m\} \models F \upharpoonright_{\omega} \cup \{f \upharpoonright_{\omega} < f\}$$

Only consider F — no need to show that any $C_i \upharpoonright_{\omega}$ implied!

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Further extensions:

- Define dominance rule w.r.t. order independent of objective
- Switch between different orders in same proof
- See [BGMN23, ABB⁺26b] for more details

Strengthening Rules: Syntax

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red ⟨Constraint C⟩ : ⟨var1⟩ -> ⟨val1⟩ . . . ⟨varN⟩ -> ⟨valN⟩ : subproof
  subproofs for proof goals
qed;
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- Witness ω should be explicitly specified in proof log
- Subproofs of proof goals should also be explicit
- Except can be skipped for proof goals that “obviously” follow, e.g.,
 - by reverse unit propagation (RUP)
 - by simple syntactic implication from other constraint (as in our reification example)
 - if the proof goal is not affected by the witness substitution ω

Successful Applications of Pseudo-Boolean Proof Logging with VERIPB

Surprisingly, pseudo-Boolean reasoning with strengthening rules sufficient to efficiently certify wide range of combinatorial solving techniques and paradigms:

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- 1 **Boolean satisfiability (SAT) solving** including advanced techniques such as
 - Gaussian elimination [GN21]
 - symmetry breaking [BGMN23, ABB⁺26b]
- 2 **SAT-based optimisation (MaxSAT)** [VDB22, BBN⁺23, BBN⁺24, IOT⁺24, VCB26]
- 3 **Pseudo-Boolean solving and optimization** [GMNO22, KLM⁺25]
- 4 **Graph solving** [GMN20, GMM⁺20, GMM⁺24, DKK⁺26]
- 5 **Dynamic programming and decision diagrams** [DMM⁺24]
- 6 **Presolving in 0–1 integer linear programming** [HOGN24]
- 7 **Constraint programming** [EGMN20, GMN22, MM23, MMN24, MM25]
- 8 **Automated planning** [DHN⁺25]
- 9 **Enumeration of solutions/models** [MNOT26]

VERIPB Resources

VERIPB [tutorials, technical documentation, and more information](#) at
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Lots of [concrete example files](#) at <https://gitlab.com/MIAOresearch/software/VeriPB>

Summing up the First Lecture



- Combinatorial solving and optimisation is a true success story
- But ensuring correctness is a crucial, and not yet satisfactorily addressed, concern
- Certifying solvers producing machine-verifiable proofs of correctness seems like most promising approach
- Cutting planes reasoning with pseudo-Boolean constraints seems to hit a sweet spot between simplicity and expressivity

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Questions so far?

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